EmbMerge: A Transformer-based Method for Fusing CDR Lists

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Introduction

- **The Problem**: Modern recommendation systems suffer from data sparsity and cold-start problems.
- **The Cross-Domain (CDR) Solution:** CDR methods transfer knowledge from a data-rich source domain to a sparse target domain.
- **The Challenge:** Different CDR models have varying architectures and capture unique, complementary signals. No single model perfectly captures complete user preferences.
- **Limitation of Existing Fusion:** Traditional rank fusion techniques rely on ranks or sparse, hand-crafted statistical features that fail to capture deep semantic relationships between items.

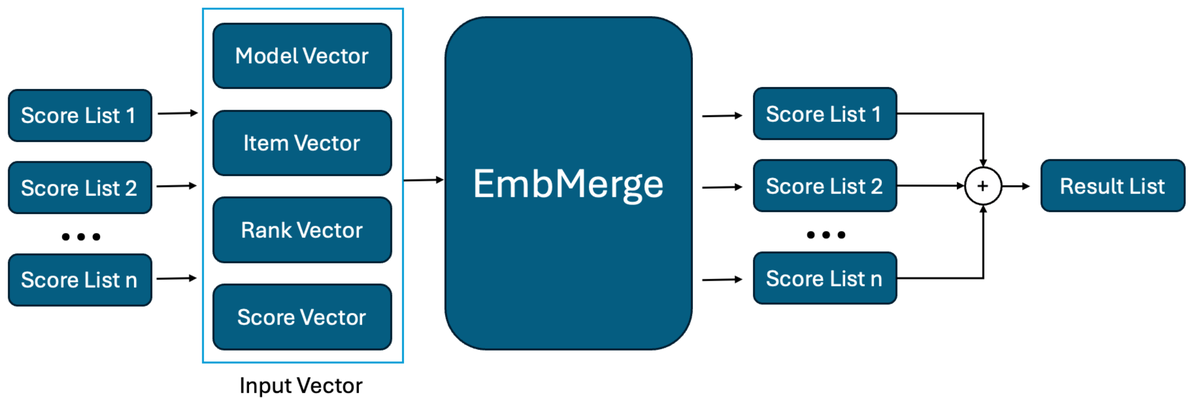
EmbMerge

A supervised rank fusion method using dense embeddings and a Transformer encoder to fuse multiple CDR lists into a single, superior ranking. We propose two different variants: Standard and Direct Prediction Models.

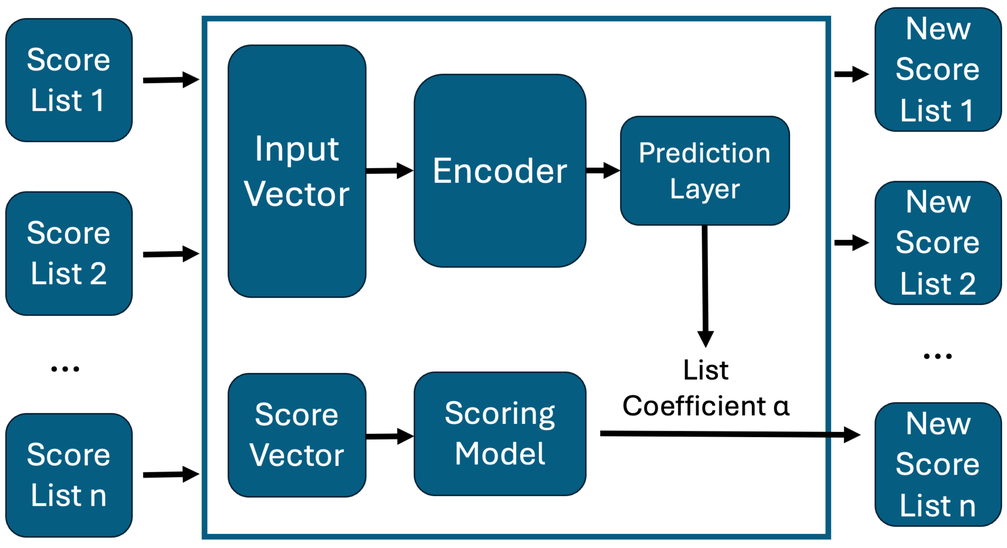
Input Representation
Instead of sparse features, we concatenate four dense vectors per item:

- **Model Vector**: Source CDR model.
- **Item Vector**: Unique item characteristics.
- **Rank Vector**: Position in the ranked list (e.g., 1–100).
- **Score Vector**: Normalized raw score.

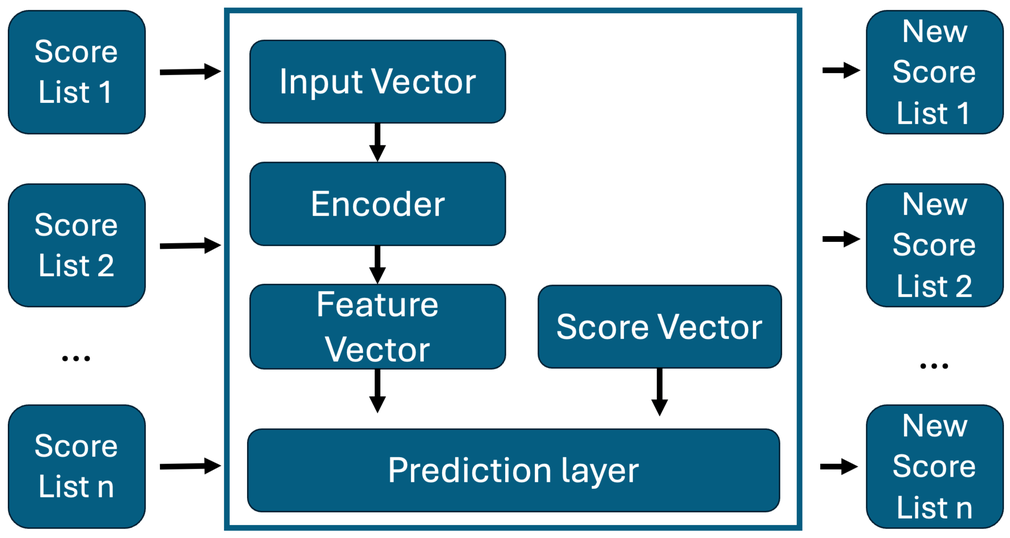
Architecture



Standard Model



Direct Prediction Model



Experiments

Datasets:

- **Digix:** Source: News Feeds → Target: Ads (439k test lists)
- **Amazon:** Source: Books → Target: Movies & TV Shows (130k test lists)

Base CDR Models: BiTGCF, CMF, and DTCDR. (1 positive and 99 negative items per list)

Baselines for Comparison: Unsupervised (RRF, BordaFuse) and Supervised (Logn-ISR, W-BordaFuse)

	Digix		Amazon	
Methods / Metrics	**Recall@10**	**NDCG@10**	**Recall@10**	**NDCG@10**
Base CDR Models				
BiTGCF	0.8904	0.7129	0.6242	0.385
CMF	0.8613	0.6729	0.6016	0.3853
DTCDR	0.878	0.6483	0.5739	0.3423
Fusion Baselines				
W-BordaFuse	0.9038	0.7245	0.6541	0.4134
BordaFuse	0.9011	0.7191	0.6519	0.4095
Logn-ISR	0.8953	0.7086	0.6469	0.4062
RRF	0.9026	0.7187	0.6595	0.4145
EmbMerge (Ours)				
Standard Model	0.9007	0.7237	**0.6619**	**0.4215**
Direct Prediction	**0.9196**	**0.7279**	0.6459	0.4035

Conclusion

Either one of the EmbMerge variants consistently outperforms individual CDR models and strong supervised/unsupervised fusion baselines across different datasets.

Replacing sparse statistical features with learned, dense embeddings successfully captures the complex, deep-semantic relationships between models, items, ranks and scores.

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